

A New Analysis of the Morphological Structure of Japanese Nominal Adjectives¹

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Abstract:

The present paper reanalyses the morphological structure of Japanese nominal adjectives (NAs) — also referred to as *na*-adjectives — in line with the author's preprint: Makino, S. (2025). *A New Analysis of the Morphological Structure of Japanese Verbal Adjectives*. Preprint, LingBuzz/Zenodo. <https://doi.org/10.5281/zenodo.15764311>. This is a preliminary version, and further revisions and clarifications will follow.

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1. Introduction

Japanese nominal adjectives (NAs) are often analysed, in parallel with "NPs + the copula =*da*", as "so-called NA stems + the copula =*da*" — e.g. *gakusei=da* "it is a student" and *sizuka=da* "it is quiet/silent." However, this analysis has certain limitations, such as the fact that in the latter there exist inflected forms ending in *-ni* — e.g. *sizuka-ni* — whereas in the former there are none, since *gakusei=ni* is a postpositional phrase headed by the postposition =*ni*. The present paper argues that **it is precisely this *-ni* which constitutes the defining constituent of NAs**, and undertakes an analysis of their morphological structure. (from the perspective of Distributed Morphology, this constituent may be regarded as an *overt* instantiation of the categorial head *a*).

2. The identification of the NA stem, NA radical and NA thematic suffix

The verbal-adjective stem (VA stem) *taka-ku* from *takai* "to be high/tall", composed of the verbal-adjective radical (VA radical) *taka-* plus the verbal-adjective thematic suffix (VA thematic suffix) *-ku*, and *sizuka-ni* from the nominal adjective (NA) *sizukada* "to be quiet/silent", constituted by the so-called nominal-adjective stem *sizuka-* plus the postposition *-ni*, can occur in the same syntactic-semantic environments, as shown in (1) – (4), where "-" and "=" indicate affixation and enclisis, respectively:

- (1) *Kare=wa taka-ku mat-ta*. "He danced high."
- (2) *Kare=wa sizuka-ni mat-ta*. "He danced quietly."
- (3) *taka-ku nar-u / taka-ku su-ru* "to become/make high/tall"
- (4) *sizuka-ni nar-u / sizuka-ni su-ru* "to become/make quiet/silent"

In (1) and (2), both *taka-ku* and *sizuka-ni* function as adverbial adjuncts of the verbal predicate *mat-* — an allomorph of the stem *maw-* from the consonantal-stem verb *ma(w)-u* "to dance" — and qualify the manner of dancing, while in (3) and (4) both serve as complements of the verbal predicates *nar-* and *su-* from the state-change verbs *nar-u* "to become" and *su-ru* "to make", denoting the results of certain state changes. (Note that the inflexional endings *-ru* $\approx$ *-u*, specified as [–past], and *-ta*, specified as [+past], are tense suffixes and function as the heads of

tense phrases at the sentential level, independently of verbal stems such as *maw-*, *nar-* and *su-* cited above.) It is thus evident that in these examples both *taka-ku* and *sizuka-ni* occur in the same phrasal positions and fulfil the same semantic function. If *taka-ku* as a whole functions as a VA stem, it can be assumed that *sizuka-ni* as a whole functions in parallel as a nominal adjective stem — hereafter, **NA stem**.

Just as, in the VA stem *taka-ku*, *taka-* and *-ku* function as VA radical and VA thematic suffix, respectively, it can be assumed that, in the NA stem *sizuka-ni*, the so-called nominal-adjective stem *sizuka-* in fact constitutes a nominal-adjective radical — hereafter, **NA radical** — which encodes the lexical meaning specific to the nominal adjective, and that *-ni*, derived from the postposition *=ni* as in *Tokyo=ni* "in/to/at Tokyo", functions as the **NA thematic suffix** — namely, a **VA stem-forming suffix** — and licenses the NA radical as a syntactic constituent within the NA category at the sentential level. Thus, the NA radical and the NA thematic suffix are the two immediate constituents of the NA stem as indicated in (5), where "-" indicates suffixation:

(5)

	Stem	
	Radical	Thematic suffix
VA	<i>taka-</i>	<i>-ku</i>
NA	<i>sizuka-</i>	<i>-ni</i>

Thus, just as the VA radical *taka-* alone, lacking the VA thematic suffix *-ku*, cannot function as a constituent of the sentence, so too the NA radical *sizuka-* alone, lacking the NA thematic suffix *-ni*, cannot function as a constituent of the sentence, as indicated in (1) – (4) and (6) – (9):

(6) **Kare=wa taka mat-ta*. "He danced high." (cf. *taka-ku mat-ta*)

(7) **Kare=wa sizuka mat-ta*. "He danced quietly." (cf. *sizuka-ni mat-ta*)

(8) **taka nar-u* / **taka su-ru* "to become/make high/tall" (cf. *taka-ku nar-u* / *su-ru*)

(9) **sizuka nar-u* / **sizuka su-ru* "to become/make quiet/silent" (cf. *sizuka-ni nar-u* / *su-ru*)

On the other hand, there is a fundamental difference between the **inflected form ending in the NA thematic suffix *-ni*** and the **postpositional phrase headed by the postposition *=ni* with an NP complement**: the former can function as an adverbial adjunct of predicates, whereas the latter cannot function as such but rather as an argument of the predicate as shown in (10) – (12):

(10) *Kare=wa sizuka-ni arui-ta.* "He walked quietly."

(11) **Kare=wa yūzin=ni arui-ta.* "*He walked to a friend."

(12) *Kare=wa yūzin=ni hon=o kasi-ta.* "He lent a book to a friend."

Therefore, it seems plausible to distinguish functionally the NA thematic suffix *-ni* from the postposition *=ni*: the former, as the head of stems, merges with NA radicals — e.g. *sizuka-*, *odayaka-*, *kantan-* — to form stems of the NA category, whereas the latter, as the head of phrases, merges with NPs — e.g. *yūzin*, *taisetuna yūzin*, *mukasi=kara=no yūzin*, etc. — to form postpositional phrases. (Note that in this study it is assumed that a phonologically zero AdvP head merges with VAPs or NAPs — e.g. *takaku-ku*, *sizuka-ni* — to form AdvPs, which function as adjuncts of predicates.) The form *siawase*, which can function both as an NA radical and as an NP, forms an NA stem when merged with the NA thematic suffix *-ni*, but forms a PP when merged with the postposition *=ni*, as indicated in (13) and (14):

(13) *Kare=wa kazoku=to siawase-ni kurasite=iru.* "He is living happily with his family."

(14) *Kono miti=wa siawase=ni tuduite=iru.* "This road leads to happiness."

It can be concluded from these observations that the **NA thematic suffix *-ni* functions morphosyntactically differently from the formally identical postposition *=ni*** and as an **indispensable morphological constituent for the construction of the NA stem**.

3. The sequence of the NA thematic suffix *-ni* and the endings *-te*, *-temo*, *-tatte*

The major difference between the NA stem *sizuka-ni* and the VA stem *taka-ku* is that the former cannot occur alone at the end of non-sentence-final coordinated clauses, as shown in (15) and (16):

(15) *Kanozyo=wa se=ga **taka-ku**, sosite utukusii/utukusikatta.* "She is/was tall and beautiful."

(16) **Kare=wa **sizuka-ni**, sosite odayakada/odayakadatta.* "He is/was silent and calm."

Nevertheless, the inflected form *sizuka-de* "to be quiet/silent and", corresponding semantically to the VA inflected form *taka-ku-te* "to be high/tall and" — constituted by the VA stem *taka-ku* followed by the atemporal conjunctive suffix *-te* specified as [0past] — is able to occur in this position, as indicated in (17) and (18):

(17) *Kanozyo=wa se=ga taka-ku-te, utukusii/utukusikatta*. "She is/was tall and beautiful."

(18) *Kare=wa sizuka-de, odayakada/odayakadatta*. "He is/was silent and calm."

From the equation *taka-ku* : *taka-ku-te* = *sizuka-ni* : *suzuka-de*, it can be hypothesised that **-de** is a **surface form of the underlying sequence -ni-te**. In reality, in PPs — postpositional phrases — one can observe free variation between the postposition =*de* "in, at" and the postposition-plus-suffix sequence =*ni-te* "idem.", as in *Tokyo=de* ~ *Tokyo=ni-te* "in/at Tokyo", though the latter is a more formal and literal form ("=" indicates enclisis). Furthermore, the assumed underlying form *sizuka-ni-te* is historically attested as an actual classical form, which constitutes the etymon of the modern *sizuka-de*. It does not seem implausible, therefore, to assume that the surface form *suzuka-de* derives synchronically from the underlying *sizuka-ni-te*, on the condition that in contemporary language the surface realisation of /-ni-te/ as [-de] is obligatory in NA stems. Building on this interpretation, the forms *sizuka-demo* and *sizuka-datte* can be analysed in the same way as *sizuka-de*, as in (19):

(19)	Stem		Ending	Surface form
	Radical	Thematic suffix		
NA	<i>sizuka-</i>	<i>-ni</i>		<i>sizuka-ni</i>
	<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>sizuka-de</i>
	<i>sizuka-</i>	<i>-ni</i>	<i>-temo</i>	<i>sizuka-demo</i>
	<i>sizuka-</i>	<i>-ni</i>	<i>-tatte</i>	<i>sizuka-datte</i>
VA	<i>taka-</i>	<i>-ku</i>		<i>taka-ku</i>
	<i>taka-</i>	<i>-ku</i>	<i>-te</i>	<i>taka-ku-te</i>
	<i>taka-</i>	<i>-ku</i>	<i>-temo</i>	<i>taka-ku-temo</i>
	<i>taka-</i>	<i>-ku</i>	<i>-tatte</i>	<i>taka-ku-tatte</i>

The problem with this analysis is that the shaded derivation *sizuka-ni-tatte* → *sizuka-datte* is neither diachronically attested nor consistent with our native-speaker intuition. In this paper, nevertheless, for the reasons set out in Section 5, this analysis is maintained.

4. The analysis of the past form and morphosyntactically related forms

The past form of the NA *sizukada* exhibits free variation between *sizuka-de#at-ta* and *sizuka-d+at-ta*, both ending in the tense suffix *-ta* specified as [+past]. The former is more formal and literal than the latter, which derives formally from the former through the deletion of the vowel *-e* in *+de* before the following vowel *a-* in *at-*. In the former *sizuka+de#at-ta*, it is possible to insert at the sentential level the focus particle *=mo* "too, also" or the topicalisation/contrast particle *=wa* "as for, etc." — both of which are enclitics — immediately after *sizuka-de*, without insertion of new constituents or deletion of existing ones, as in (20), where "#" denotes a morphological word boundary that coincides with a prosodic word boundary:

(20) *sizuka-de#at-ta* → *sizuka-de=mo#atta* / *sizuka-de=wa#at-ta*

In *sizuka-de#at-ta*, therefore, both *sizuka-de* and *at-ta* constitute morphologically autonomous words. If the *-de* in *sizuka-de#at-ta* derives from *-ni-te*, much like the *-de* in *sizuka-de* used alone as an atemporal form as in (18), and since *at-* is the allomorph of the stem *ar-* of the verb *ar-u* "to be", the derivation from the underlying form — henceforth, UF — to the surface forms — hereafter, SFs — can be assumed as schematised in (21):

(21) UF: *sizuka-ni-te#ar-tar* → SF: *sizuka-de#at-ta* → SF: *sizuka-d+at-ta*

This derivation proceeds in almost exactly the same manner as that of the past form of VAs such as *taka-k+at-ta* from *takai* in (22), where IF represents an intermediate form:

(22) UF: *taka-ku#ar-tar* → IF: *taka-ku#at-ta* → SF: *taka-k+at-ta*

Just as the VA thematic suffix *-ku* deletes its vowel *-u* before the light verb stem *at-*, the suffix sequence *-de*, derived from *-ni-te*, deletes its vowel *-e* in the same morphosyntactic context. The only difference is that *takaku* in (22) is a **stem** without ending, whereas *sizuka-ni-te*, realised as *sizuka-de*, constitutes a **word** composed of the stem *sizuka-ni* and an inflexional suffix *-te*.

The reason why, in the derivation *sizuka-ni-te#ar-tar* → *sizuka-de#at-ta* → *sizuka-d+at-ta* in (21), there appears stem allomorphs *ar-* and *at-* of the light verb *ar-u* — is that the inflexional

suffix *-ta* cannot combine with the same type of inflexional suffix *-te*, as in the hypothetical **sizuka-ni-te-ta*. Consequently, to resolve this suffix sequence, the stem *ar-* of the light verb *ar-u* is inserted before *-ta*, with which it constitutes a new word *ar-tar*, realised as *at-ta*. The deletion of the stem-final *-e* in the surface *sizuka-de* before *at-* in (21) seems to indicate the compounding of *sizuka-de* with *at-*, much like that of *-u* in *taka-ku* before the same *at-* in (22).

Under this analysis, the morphological structure of *sizuka-de#at-ta* and *sizuka-d+at-ta* in (21) may be represented as in (23), where "+" indicates compounding, and "them.suff." and "inflex.suff." abbreviate "thematic suffix" and "inflexional suffix", respectively:

(23)

Word (when <i>-ni-te</i> → <i>-d</i>)					Surface form
Word			Word		
Stem		Ending	Light verb stem	Ending	
Radical	Them.suff.	Inflex.suff.	Light verb radical	Inflex. suff.	
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar [+past]</i>	
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar [+past]</i>	
					<i>sizuka-d+at-ta</i>
					<i>sizuka-de#at-ta</i>

Other forms of this type with verbal endings *-tar-ō*, *-tar-a(-ba)*, *-tar-i*, *-ru*, *-reba*, *-jō*, *-e*, *-tutu* and *-nagara* — including the forms in (19) — can be analysed as shown in (24):

(24)

Word (when <i>-ni</i> , <i>-ni-te</i> → <i>-n</i> , <i>-d</i>)				
Word			Word	
Stem		Ending	LV stem	Ending
Radical	Them.suff.	Inflex.suff.	LV radical	Inflex.suff.
<i>sizuka-</i>	<i>-ni</i>			
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>		
<i>sizuka-</i>	<i>-ni</i>	<i>-temo</i>		
<i>sizuka-</i>	<i>-ni</i>	<i>-tatte</i>		
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-ru</i> [−past]
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-reba</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-jō</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-jō</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-e</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar</i> [+past]
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar</i> [+past]
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-ō</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-ō</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-a(-ba)</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-a(-ba)</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-i</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-i</i>

Surface form
<i>sizuka-ni</i>
<i>sizuka-de</i>
<i>sizuka-demo</i>
<i>sizuka-datte</i>
<i>sizuka-de#ar-u</i>
<i>sizuka-de#ar-eba</i>
<i>sizuka-d+ar-ō</i>
<i>sizuka-de#ar-ō</i>
<i>sizuka-de#ar-e</i>
<i>sizuka-d+at-ta</i>
<i>sizuka-de#at-ta</i>
<i>sizuka-d+at-tar-ō</i>
<i>sizuka-de#at-tar-ō</i>
<i>sizuka-d+at-tar-a(-ba)</i>
<i>sizuka-de#at-tar-a(-ba)</i>
<i>sizuka-d+at-tar-i</i>
<i>sizuka-de#at-tar-i</i>

<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tutu</i>	<i>sizuka-de#ari-tutu</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-nagara</i>	<i>sizuka-de#ari-nagara</i>

In the underlying forms of *sizuka-de* and *sizuka-demo* — namely, *sizuka-ni-te* and *sizuka-ni-temo* — it is possible to insert the particles =*mo* or =*wa* at the sentential level between the NA stem *sizuka-ni* and the inflexional suffixes *-te* and *-temo*, just as it is between the VA stem *taka-ku* and the same suffixes *-te* and *-temo* in *taka-ku-te* and *taka-ku-temo*, as shown in (25) – (28):

- (25) *sizuka-ni* + =*wa* + *-te* → *sizuka-ni-te*=*wa#ar-te* → *sizuka-de*=*wa#at-te*
- (26) *taka-ku* + =*wa* + *-te* → *taka-ku*=*wa#ar-te* → *taka-ku*=*wa#at-te*
- (27) *sizuka-ni* + =*wa* + *-temo* → *sizuka-ni-te*=*wa#ar-temo* → *sizuka-de*=*wa#at-temo*
- (28) *taka-ku* + =*wa* + *-temo* → *taka-ku*=*wa#ar-temo* → *taka-ku*=*wa#at-temo*

In this operation, apart from the fact that the addition of *-te* to the NA stem *sizuka-ni* yields the word-level form *sizuka-ni-te*, which surfaces as *sizuka-de*, the morphosyntactic process is identical: through the insertion of =*wa*, the suffixes *-te* and *-temo* are separated from the stems *sizuka-ni* and *taka-ku* and thereby rendered stemless, whereupon the stem *ar-* of the light verb *ar-u* is inserted before them, resolving this stemlessness. This proves that *sizuka-ni* and *taka-ku* alike function as predicative stems. This operation can also be applied to the underlying form of the NA *sizuka-datte* — namely, *sizuka-ni-tatte* hypothesised in (19) — as well as to the VA *taka-ku-tatte*, as indicated in (29) and (30):

- (29) *sizuka-ni* + =*wa* + *-tatte* → *sizuka-ni-te*=*wa#ar-tatte* → *sizuka-de*=*wa#at-tatte*
- (30) *taka-ku* + =*wa* + *-tatte* → *taka-ku*=*wa#ar-tatte* → *taka-ku*=*wa#at-tatte*

We posit *sizuka-ni-tatte* as the UF of *sizuka-datte* — which is neither diachronically attested nor consistent with our native-speaker intuition — on the basis of the homogeneity of the processes in (25), (27) and (29).

5. The other forms: *sizuka-d+a*, *sizuka-n+a*, *sizuka-n+ar-u*, *sizuka-n+ar-e*

The surface non-past form shows free variation between *sizuka-de#ar-u* and *sizuka-d+a*, the former being more formal and literal. It can be observed that, in the latter, the vowel *-e* in the suffix sequence *-de* is deleted before another vowel *a-*, much like the *-e* of *-de* in *sizuka-de#at-ta* → *sizuka-d+at-ta* discussed in Section 4. Furthermore, neither the final consonant *-r* of the light verb stem *ar-* nor the allomorph *-u* of the non-past suffix *-ru* appear. The more informal and colloquial form *sizuka-d+a* nevertheless has the same non-past value as *sizuka-de#ar-u*. From this, it can be hypothesised that, in *sizuka-d+a*, the non-past suffix does not bear the overt form *-ru* but the phonologically zero *-0*, just as it does in VA non-past forms such as *taka-i-0*, and that the final *-r* of the light verb stem is deleted at the surface level by virtue of its position at the word-final boundary. Thus, the derivation from the UF to the SF via the IF can be assumed as in (31):

(31) UF: *sizuka-ni-te#ar-0* → IF: *sizuka-de#a* → SF: *sizuka-d+a*

Historically, although a copula, the form *=de#a*, which yielded the modern *=d+a*, can be attested in Azuchi–Momoyama-period materials: *bonnin yorimo giūzaini fuxōzuru cotodea* "(it) is a reason for imposing a heavier guilt (on him) than on an ordinary person" (Amakusa Edition of *Esopo no fabulas*, 1593, p.475, ll.10-11, in Text Data Sets for Research on the History of Japanese). The attributive form exhibits practically the same type of free variation between *sizuka-n+ar-u* and *sizuka-n+a* as *sizuka-de#ar-u* and *sizuka-d+a*, the former being more formal and literal. In these forms, *sizuka-n-* can be interpreted as the underlying NA stem *sizuka-ni* deleting its final vowel *-i* before the following vowel *a-*. (Note that, historically, the etymon of *n+ar-u* was in fact *=ni#ar-u*). Thus, the derivation of each form from the UF to the SF can be assumed as in (32) and (33), respectively:

(32) UF: *sizuka-ni#ar-ru* → SF: *sizuka-n+ar-u*

(33) UF: *sizuka-ni#ar-0* → SF: *sizuka-n+a*

The morphological structure of the forms in (31) – (33) — including the imperative form *sizuka-n+ar-e* "Be quiet/silent", which is more formal and literal than the current *sizuka-de#ar-*

e, and the conditional form *sizuka-n+ar-a(-ba)* "if it is quiet/silent", with a semantically synonymous yet morphologically distinct form *sizuka-de#ar-eba* — is represented in (34):

(34)

Word (when <i>-ni</i> , <i>-ni-te-</i> → <i>-n-</i> , <i>-d</i>)					Surface form
Word			Word		
Stem		Ending	LV stem	Ending	
Radical	Them.Suff.	Inflex.Suff.	LV radical	Inflex.Suff.	
<i>sizuka-</i>	<i>-ni</i>		<i>ar-</i>	<i>-0 [-past]</i>	
<i>sizuka-</i>	<i>-ni</i>		<i>ar-</i>	<i>-ru [-past]</i>	
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-0 [-past]</i>	
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-ru [-past]</i>	
<i>sizuka-</i>	<i>-ni</i>		<i>ar-</i>	<i>-e</i>	
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-e</i>	
<i>sizuka-</i>	<i>-ni</i>		<i>ar-</i>	<i>-a(-ba)</i>	
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-reba</i>	

<i>sizuka-n+a</i>
<i>sizuka-n+ar-u</i>
<i>sizuka-d+a</i>
<i>sizuka-de#ar-u</i>
<i>sizuka-n+ar-e</i>
<i>sizuka-de#ar-e</i>
<i>sizuka-n+ar-a(-ba)</i>
<i>sizuka-de#ar-eba</i>

6. NAs ending in *-d+a* /*-ni-te#ar-0/* vs. the copula *=d+a* /*=de#ar-0/*

NAs such as *sizuka-d+a* have often been analysed — in parallel with the structure "NPs + the copula *=da*", such as *gakusei* + *=da* — as "so-called NA stems + the copula *=da*", such as *sizuka* + *=da*. However, there are notable differences between these two structures, as noted in (35) — (37):

(35) As stated in Section 2, NAs have an inflected form ending in *-ni* — e.g. *sizuka-ni* — whereas NPs + *=da* do not have an inflected form consisting of NP + *=ni*. Such forms — e.g. *gakusei=ni* — are syntactically PPs.

(36) In contemporary standard Japanese, NAs exhibit inflected forms that contain *-ni* but not *-te*, such as *sizuka-n+a daiti* "a land that is quiet", *sizuka-n+ar-u daiti* "idem." (more formal and literal) and *sizuka-n+ar-e* "be quiet/silent" (more formal and literal). By contrast, NPs + *=da* do not generally exhibit inflected forms with *-ni* but without *-te*. Forms such as **gakusei=n+a wakamono* "a youth that is a student", **gakusei=n+ar-u wakamono* "idem." and **gakusei=n+ar-e* "be a student" — which would be *gakusei=no wakamono*, *gakusei=de#ar-u wakamono* and *gakusei=de#ar-e* in standard language — are not normally used.

(37) In certain fixed expressions such as "NPs + *=n+a mono=de*" and "NPs + *=n+ar-u mono*"

— e.g. *watasi=wa gakusei=n+a mono=de* "because/since I am (qualified/characterised as) a student" and *gakusei=n+ar-u=mono* "that which is (qualified/characterised as) a student" — and in colloquial expressions such as *gakusei=n+a kanzi*, which is nearly synonymous with *gakuseiteki=n+a kanzi* "a student-like impression", inflected forms with *-ni* but without *-te* can be used. In these examples, nevertheless, **the NP *gakusei* does not seem to refer to an individual but rather to properties, qualities or characteristics, and thus is semantically closer to NA radicals.** (The only exception is, to our knowledge, *gakusei=n+ar-a(-ba)* "if it is a student", in which the NP *gakusei* refers to an individual. This fixed conditional expression *=n+ar-a(-ba)*, derived from the underlying *=ni#ar-a(-ba)*, appears to be a fossilised form inherited from classical Japanese, where *=n+ari*, derived from the underlying *=ni#ar-0*, functioned as the standard copula.)

If the *-da* of NAs is regarded as the same copula as *=da* in NPs + *=da*, it becomes difficult to account for the presence or absence of inflected forms ending in *-ni* in (35), the presence or absence of forms containing only *-ni* in contemporary standard language in (36) and the affinity of *-ni* with NPs that denote not individuals but properties, qualities or characteristics in (37).

By contrast, if it is assumed that, as stated in Section 2, ***-ni* is an indispensable morphological constituent for the construction of the NA stem**, whereas the copula *=da*, which merges with NPs, does not analytically contain this *-ni* — in other words, that the minimal underlying unit involved in the copula *=da* is not *=ni-te* but rather *=de* — then the facts become explicable. From these observations, it can be inferred that **the copula *-da***, which merges with NPs, can be analysed as **the copula stem *=de* + the stem *ar-* of the light verb *ar-u* + inflexional endings**, whereas **NA inflected forms** can be analysed as **the NA radical + the NA thematic suffix(es) *-ni(-te)* + the stem *ar-* of the light verb *ar-u* + inflexional endings**.

Futhermore, comparing *sizuka-n+ar-u koto* with *sizuka-de#ar-u koto*, it must be concluded that *-ni* and *-de* do not exhibit complementary distribution, since both occur in the same underlying context *sizka-#ar-ru*, as in *sizuka-ni#ar-ru* ($\rightarrow$ *sizuka-n+ar-u*) and *sizka-ni-te#ar-ru* ($\rightarrow$ *sizuka-de#ar-u*). Accordingly, it cannot be concluded that *-ni* and *-de* are merely allomorphs of a single thematic suffix in NA stems, the latter having to be analysed underlyingly as *-ni-te*.

7. Isomorphism between NA inflected forms and periphrastic V forms in *-te#ar-u*

The NA inflected form *sizuka-de#at-ta* has exactly the same pattern of underlying structure as, for example, the periphrastic verbal form that denotes a result state, *nob-e-te#at-ta* "it had extended/stated", derived from the underlying *nob-e-te#ar-tar*, as indicated in (38) and (39):

(38) UF: *sizuka-ni-te#ar-tar* → SF: *sizuka-de#at-ta* → SF: *sizuka-d+at-ta*

(39) UF: *nob-e-te#ar-tar* → SF: *nob-e-te#at-ta*

The common structure shared by both forms is: [[[radical] thematic suffix] inflexional suffix *-te*] + [[[light verb radical *ar-*] inflexional suffix(es)], as schematised in (40):

(40)	Word			Word		Surface form
	Stem		Ending	LV stem	Ending	
	Radical	Them.suff.	Inflex.suff.	LV radical	Inflex.suff.	
	<i>nob-</i>	<i>-e</i>				<i>nob-e</i>
	<i>sizuka-</i>	<i>-ni</i>				<i>sizuka-ni</i>
	<i>nob-</i>	<i>-e</i>	<i>-te</i>			<i>nob-e-te</i>
	<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>			<i>sizuka-de</i>
	<i>nob-</i>	<i>-e</i>	<i>-temo</i>			<i>nob-e-temo</i>
	<i>sizuka-</i>	<i>-ni</i>	<i>-temo</i>			<i>sizuka-demo</i>
	<i>nob-</i>	<i>-e</i>	<i>-tatte</i>			<i>nob-e-tatte</i>
	<i>sizuka-</i>	<i>-ni</i>	<i>-tatte</i>			<i>sizuka-datte</i>
	<i>nob-</i>	<i>-e</i>	<i>-te</i>	<i>ar-</i>	<i>-ru</i> [–past]	<i>nob-e-te#ar-u</i>
	<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-ru</i> [–past]	<i>sizuka-de#ar-u</i>
	<i>nob-</i>	<i>-e</i>	<i>-te</i>	<i>ar-</i>	<i>-reba</i>	<i>nob-e-te#ar-eba</i>
	<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-reba</i>	<i>sizuka-de#ar-eba</i>
	<i>nob-</i>	<i>-e</i>	<i>-te</i>	<i>ar-</i>	<i>-jō</i>	<i>nob-e-te#ar-ō</i>
	<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-jō</i>	<i>sizuka-de#ar-ō</i>
	<i>nob-</i>	<i>-e</i>	<i>-te</i>	<i>ar-</i>	<i>-tar</i> [+past]	<i>nob-e-te#at-ta</i>
	<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar</i> [+past]	<i>sizuka-de#at-ta</i>
	<i>nob-</i>	<i>-e</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-ō</i>	<i>nob-e-te#at-tar-ō</i>
	<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-ō</i>	<i>sizuka-de#at-tar-ō</i>
	<i>nob-</i>	<i>-e</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-a(-ba)</i>	<i>nob-e-te#at-tar-a(-ba)</i>
	<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-a(-ba)</i>	<i>sizuka-de#at-tar-a(-ba)</i>
	<i>nob-</i>	<i>-e</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-i</i>	<i>nob-e-te#at-tar-i</i>
	<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-i</i>	<i>sizuka-de#at-tar-i</i>
	<i>nob-</i>	<i>-e</i>	<i>-te</i>	<i>ar-</i>	<i>-tutu</i>	<i>nob-e-te#ari-tutu</i>
	<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tutu</i>	<i>sizuka-de#ari-tutu</i>
	<i>nob-</i>	<i>-e</i>	<i>-te</i>	<i>ar-</i>	<i>-nagara</i>	<i>nob-e-te#ari-nagara</i>
	<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-nagara</i>	<i>sizuka-de#ari-nagara</i>

The only fundamental difference is that the thematic suffix is *-ni* in NA stems — e.g. *sizuka-*

ni-te#ar-ru — whereas in V stems it is *-e* or *-i* in vowel-stem verbs — e.g. nob-e-te#ar-ru, n-i-te#ar-ru "it has boiled" — and *-0* in consonant-stem verbs — e.g. yom-0-te#ar-ru "it has read", realised as yom-de#ar-u. (Note moreover that in the V stem + *-te#ar-ru* construction, compounding of the type *sizuka-de#at-ta* → *sizuka-d+at-ta* does not occur.) This means that **nominal adjectives, like true verbs and verbal adjectives, belong to a category that inflects verbally**, and that their morphological structure cannot be adequately analysed as "NA + the copula =*da*" without leaving serious doubts, as already stated in the previous section.

8. Conclusion

The inflected forms of the NA *sizukada* can be morphologically analysed as shown in (41):

(41)

Word (when <i>-ni</i> , <i>-ni-te-</i> → <i>-n-</i> , <i>-d</i>)					Surface form
Word			Word		
Stem		Ending	LV stem	Ending	
Radical	Them.Suff.	Inflex.Suff.	LV radical	Inflex.Suff.	
<i>sizuka-</i>	<i>-ni</i>				<i>sizuka-ni</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>			<i>sizuka-de</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-temo</i>			<i>sizuka-demo</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-tatte</i>			<i>sizuka-datte</i>
<i>sizuka-</i>	<i>-ni</i>		<i>ar-</i>	<i>-0 [-past]</i>	<i>sizuka-n+a</i>
<i>sizuka-</i>	<i>-ni</i>		<i>ar-</i>	<i>-ru [-past]</i>	<i>sizuka-n+ar-u</i>
<i>sizuka-</i>	<i>-ni</i>		<i>ar-</i>	<i>-e</i>	<i>sizuka-n+ar-e</i>
<i>sizuka-</i>	<i>-ni</i>		<i>ar-</i>	<i>-a(-ba)</i>	<i>sizuka-n+ar-a(-ba)</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-0 [-past]</i>	<i>sizuka-d+a</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-ru [-past]</i>	<i>sizuka-de#ar-u</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-reba</i>	<i>sizuka-de#ar-eba</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-jō</i>	<i>sizuka-d+ar-ō</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-jō</i>	<i>sizuka-de#ar-ō</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-e</i>	<i>sizuka-de#ar-e</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar [+past]</i>	<i>sizuka-d+at-ta</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar [+past]</i>	<i>sizuka-de#at-ta</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-ō</i>	<i>sizuka-d+at-tar-ō</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-ō</i>	<i>sizuka-de#at-tar-ō</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-a(-ba)</i>	<i>sizuka-d+at-tar-a(-ba)</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-a(-ba)</i>	<i>sizuka-de#at-tar-a(-ba)</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-i</i>	<i>sizuka-d+at-tar-i</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tar-i</i>	<i>sizuka-de#at-tar-i</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-tutu</i>	<i>sizuka-de#ari-tutu</i>
<i>sizuka-</i>	<i>-ni</i>	<i>-te</i>	<i>ar-</i>	<i>-nagara</i>	<i>sizuka-de#ari-nagara</i>

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